Lab 13

Loops + Nested Loops

Objective:

The objective of this lab will be to learn about loops and nested loops with the help of examples and learning tasks.

Activity Outcomes:

The activities provide hands - on practice with the following topics

- Implement a while loop
- Implement a for-in loop
- Implement nested loops

Instructor Note:

As a pre-lab activity, read Chapter 6 from the textbook "Python Basics: A Practical Introduction to Python 3, 2021".

1) Useful Concepts

A loop is a block of code that gets repeated over and over again either a specified number of times or until some condition is met which is why a loop runs only a finite number of times. If we specify a condition that does not get fulfilled ever during loop iterations then it would go on forever and the program would freeze.

1. One kind of a loop is a while loop. The syntax is given below. We specify a condition in the round brackets after a while just like we do for the if statement. This loop keeps on running while the statement passed is true(hence the name while).

```
while(<condition>):
    // some code to repeat
```

2. Another kind of a loop is a for-in loop. The syntax is given below. A for-in loop iterates over a sequence(list, string or range). The variable we use after **for**, gets the value of the next item in the sequence in each iteration while the variable after **in** is a sequence which we want iterated on.

```
for                     <item_name>                    in                     <sequence_name>:
    //                  some                 code                 to                    repeat
```

3. Sometimes we would want to nest loops. This can be done as well. This will look something like this

```
for <item_name> in <sequence_name>:
    // some code to repeat
    while(<some_condition>):
        // some nested code to repeat
```

2) Solved Lab Activites

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***15***	***Low***	***CLO-7***
2	***15***	***Low***	***CLO-7***
3	***15***	***Low***	***CLO-7***
4	***15***	***Low***	***CLO-7***

Activity 1:

Python program to illustrate an while loop

Solution:

```
x = 0
while(x<4):
    print("The value of x is:",x)
    x = x + 1
```

Output

```
The value of x is: 0
The value of x is: 1
The value of x is: 2
The value of x is: 3
```

Activity 2:

Python program to illustrate an for in loop.

```
subjects = ['Maths','English','Urdu']
for subject in subjects:
    print("The name of the subject is:",subject)
```

Output

```
The name of the subject is: Maths
```

```
The name of the subject is: English
The name of the subject is: Urdu
```

Activity 3:

Python program to illustrate a nested loop.

```python
subjects = ['Maths','English','Urdu']
grades = ['A','B']
for subject in subjects:
    print("The name of the subject is:",subject)
    for grade in grades:
        print("The name of the grade is:", grade)
```

Output

```
The name of the subject is: Maths
The name of the grade is: A
The name of the grade is: B
The name of the subject is: English
The name of the grade is: A
The name of the grade is: B
The name of the subject is: Urdu
The name of the grade is: A
The name of the grade is: B
```

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1

Write a program to print all natural numbers from 1 to n.

Lab Task 2

Write a program to print all even numbers between 1 to 100.

Lab Task 3

Write a program to ask user input for a number. Check if it is a prime number. If it is, end the program but if it is not ask for the user input again till the user enters a prime number.

Lab Task 4

Write a program to print a table of numbers from 1 to 8.

Lab Task 5

Write a nested for loop that prints the following output:

```
      1
    1 2 1
  1 2 4 2 1
1 2 4 8 4 2 1
```